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Teorema de la convergencia dominada

Teorema 1 (de la convergencia dominada). *Sea* $(f_n)_{n\in\mathbb{N}}$ *una sucesión de fns medibles en* (X,Σ,μ) *tal que* $\exists \lim f_n =: f \wedge \exists g$ *integrable* $: \forall n \in \mathbb{N} : |f_n| \leqslant g$[1]

$$\implies \lim_{n\to\infty}\int |f_n - f| \,\mathrm{d}\mu = 0, \quad \textit{en particular,} \qquad \int \lim_{n\to\infty} f_n \,\mathrm{d}\mu = \lim_{n\to\infty}\int f_n \,\mathrm{d}\mu.$$

***Demostración*:** Consideramos $\left(\hat{f}_n\right)_{n\in\mathbb{N}}$ dada por $\forall n\in\mathbb{N} : \hat{f}_n = f_n - f \implies \hat{f}_n \xrightarrow{n\to\infty} 0$.

Definimos $\forall n\in\mathbb{N} : h_n := g - \left|\hat{f}_n\right| \geqslant 0 \overset{\substack{\text{Fatou}\\ \downarrow}}{\implies} \int \liminf\limits_{n\to\infty} h_n \,\mathrm{d}\mu \leqslant \liminf\limits_{n\to\infty} \int h_n \,\mathrm{d}\mu$.

Ahora, como $\liminf\limits_{n\to\infty} h_n = \lim\limits_{n\to\infty} g - \left|\hat{f}_n\right| = g$ porque $\hat{f}_n \xrightarrow{n\to\infty} 0$, entonces

$$\int g \,\mathrm{d}\mu \leqslant \liminf_{n\to\infty}\int \left(g - \left|\hat{f}_n\right|\right)\mathrm{d}\mu \overset{\substack{\text{linealidad}\\ \downarrow}}{=} \liminf_{n\to\infty}\left(\int g\,\mathrm{d}\mu - \int \left|\hat{f}_n\right| \mathrm{d}mu\right)$$
$$\implies \int g\,\mathrm{d}\mu \leqslant \liminf_{n\to\infty}\int g\,\mathrm{d}\mu - \limsup_{n\to\infty}\int\left|\hat{f}_n\right|\mathrm{d}\mu$$

Como $\liminf\limits_{n\to\infty}\int g\,\mathrm{d}\mu = \int g\,\mathrm{d}\mu \wedge g \in \mathcal{L}^1(\mu)$, podemos cancelar y nos queda

$$\cancel{\int g\,\mathrm{d}\mu} \leqslant \cancel{\int g\,\mathrm{d}\mu} - \limsup_{n\to\infty}\int\left|\hat{f}_n\right|\mathrm{d}\mu \implies \limsup_{n\to\infty}\int\left|\hat{f}_n\right|\mathrm{d}\mu \leqslant 0$$

Lo que implica que $\limsup\limits_{n\to\infty}\int |f_n - f|\,\mathrm{d}\mu = 0 \implies \lim\limits_{n\to\infty}\int |f_n - f|\,\mathrm{d}\mu = 0$.

$$\text{y como } \int |f_n - f|\,\mathrm{d}\mu = \begin{cases} \int (f_n - f)\,\mathrm{d}\mu = \int f_n\,\mathrm{d}\mu - \int f\,\mathrm{d}\mu & \text{si } f_n \geqslant f \\ \int (f - f_n)\,\mathrm{d}\mu = \int f\,\mathrm{d}\mu - \int f_n\,\mathrm{d}\mu & \text{si } f_n < f \end{cases}$$

se tiene $\lim\limits_{n\to\infty}\int |f_n - f|\,\mathrm{d}\mu = 0 \iff \lim\limits_{n\to\infty}\int f_n\,\mathrm{d}\mu = \int f\,\mathrm{d}\mu = \int \lim\limits_{n\to\infty} f_n\,\mathrm{d}\mu$. ■

Referenciado en

- `lem-convergencia-uniforme-esp-finito-imp-lp`

[1]Se dice que g "domina" a todas las f_n.

- prop-convergencia-puntual-dominada-imp-lp
- prop-fn-distribucion
- teo-inversion-transformada-fourier
- lem-convergencia-lp-traslacion
- esperanza-condicionada-sigma-algebra
- prop-fn-simples-denso-lp
- ley-debil-grandes-numeros
- teo-derivacion-bajo-el-signo-integral
- lem-transformada-fourier-continua
- obs-propiedades-dilatacion-isotropica

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